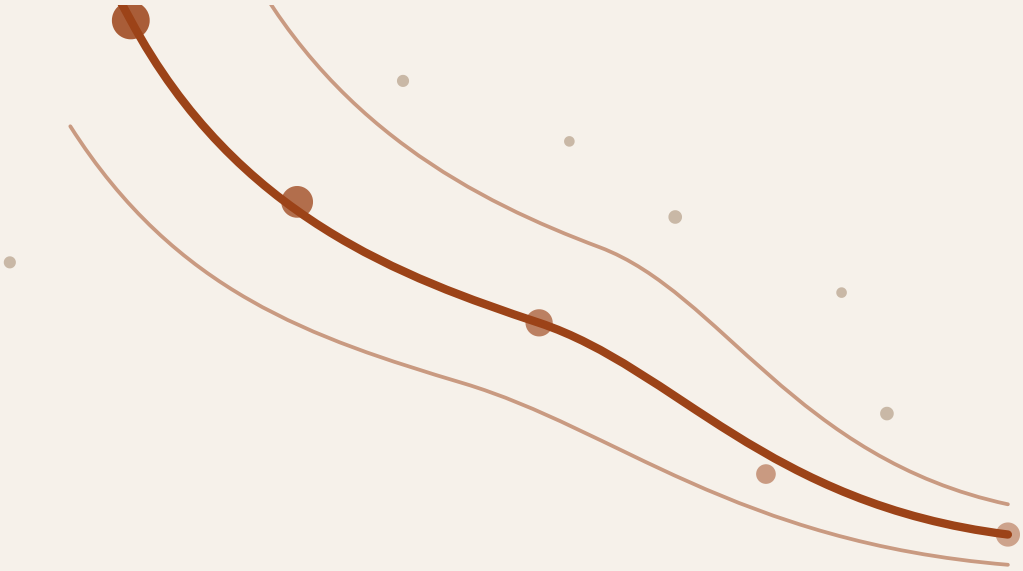
# Innovation Management

From scattered initiatives
to a sustained organizational capability

**Dr. Omar A. Alsamani**



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Back matter

# Glossary

The terms defined in the chapters of the book, with the chapter in which each definition appears. Entries are listed alphabetically.

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| **Absorptive capacity**<br>Absorptive capacity | An organization's ability to recognize the value of external knowledge, assimilate it and apply it. Chapter 18 |
| **Academic engagement**<br>Academic engagement | Faculty collaboration with external organizations through joint or contract research or consulting, as distinct from the commercialization of research. Chapter 20 |
| **Academic heartland**<br>Academic heartland | The departments and colleges where teaching and research take place, as distinct from peripheral units. Chapter 20 |
| **Architectural innovation**<br>Architectural innovation | Linking known components in a new way. Chapter 1 |
| **Boundary-spanning activities**<br>Boundary-spanning activities | A team's activities with external parties: communicating with leadership, coordinating with other units and gathering knowledge. Chapter 9 |
| **Business model**<br>Business model | The logic by which an organization delivers value to those it serves and captures enough to sustain |

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| | itself. Chapter 13 |
| **Business process innovation** Business process innovation | A new or improved change in the way an organization operates in one of its business functions. Chapter 1 |
| **Campbell's law** Campbell's law | The more a quantitative indicator is relied on in decision-making, the greater the pressure to manipulate it and to distort what it measures. Chapter 16 |
| **Certification** Certification | Verification by an independent body that an organization's system conforms to the requirements of a specified standard. Chapter 5 |
| **Challenge question** Challenge question | A formulation of a priority problem that specifies what is needed without imposing the solution. Chapter 4 |
| **Chief innovation officer** Chief innovation officer | The person who leads the orchestrating unit and achieves results through units they do not head. Chapter 6 |
| **Collingridge dilemma** Collingridge dilemma | The difficulty of controlling an innovation: it is easy to change while its effects are unknown, and hard to change once they become clear. Chapter 19 |
| **Componential model of creativity** Componential model of creativity | A model that explains individual creativity through domain-relevant skills, creativity-relevant processes and intrinsic task motivation, within the context of the work environment. Chapter 7 |
| **Concept** Concept | A description of a solution clear enough to be understood and tested, including its critical |

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| | assumptions. Chapter 12 |
| **Convergent thinking**<br>Convergent thinking | Arriving at the most suitable answer among the alternatives. Chapter 12 |
| **Creative climate**<br>Creative climate | The patterns of behavior and feeling observed in daily work that encourage or hinder creativity. Chapters 8 and 16 |
| **Creative self-efficacy**<br>Creative self-efficacy | A person's belief in their ability to produce new and useful ideas. Chapter 7 |
| **Critical assumption**<br>Critical assumption | An assumption that, if wrong, brings down the whole initiative. Chapter 14 |
| **Culture-embedding mechanisms**<br>Culture-embedding mechanisms | The practices through which people learn the actual culture, such as measurement, rewards and responses to crises. Chapter 8 |
| **Decision limbo**<br>Decision limbo | The state of an initiative that is neither accepted nor rejected, and so remains suspended without learning or impact. Chapter 5 |
| **Decision rights**<br>Decision rights | Specifying who holds the authority to make each type of innovation decision. Chapter 10 |
| **Degree of novelty**<br>Degree of novelty | The novelty of a solution relative to a unit of reference: the organization, the market or the world. Chapter 1 |
| **Design fixation**<br>Design fixation | The pull of thinking toward the features of a known solution or the first solution proposed. Chapter 12 |

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| **Disciplined collaboration**<br>Disciplined collaboration | Collaboration between units directed at a result whose value exceeds its cost. Chapter 6 |
| **Discovery-driven planning**<br>Discovery-driven planning | Planning that states assumptions explicitly and sequences the stages so that the most important are tested first. Chapter 14 |
| **Disruptive innovation**<br>Disruptive innovation | Innovation that starts at the low end of a market or in a new market, then improves until it competes with prevailing solutions. Chapter 1 |
| **Divergent thinking**<br>Divergent thinking | Producing multiple, varied answers to a single question. Chapter 12 |
| **Dominant logic**<br>Dominant logic | The way of thinking that the existing model imposes on the evaluation of new opportunities. Chapter 13 |
| **Dynamic capabilities**<br>Dynamic capabilities | An organization's ability to sense and seize opportunities and to reconfigure its resources and practices in a changing environment. Chapter 2 |
| **Ecosystem risk**<br>Ecosystem risk | The dependence of an innovation's success on partners and links in the adoption chain that the organization does not control. Chapter 18 |
| **Employee voice**<br>Employee voice | Employees' expression of ideas, suggestions and concerns with the aim of improving work. Chapter 3 |
| **Escalation of commitment**<br>Escalation of commitment | Continuing to invest in a failing course of action to justify the original decision. Chapter 10 |

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| **Executive sponsor**<br>Executive sponsor | A leader who links the initiative to strategy, gives it legitimacy and resources, and removes obstacles in its way. Chapter 3 |
| **Explainability**<br>Explainability | The ability of those using an automated system to explain the reasons behind its decisions. Chapter 19 |
| **Exploitation**<br>Exploitation | Refining and deepening what the organization already knows in order to achieve efficiency. Chapter 3 |
| **Exploration**<br>Exploration | Searching for new knowledge and opportunities under uncertainty. Chapter 3 |
| **External realities**<br>External realities | The conditions surrounding the organization and how they change, in markets, society, legislation and technology. Chapter 2 |
| **Front end of innovation**<br>Front end of innovation | The phase before a project is formally launched, including identifying needs, exploring opportunities and generating ideas. Chapter 1 |
| **Front-loading**<br>Front-loading | Identifying and solving problems in the early stages, when change is cheaper. Chapter 17 |
| **Hard core and soft periphery**<br>Hard core and soft periphery | The distinction between what must remain fixed in an innovation and what can be adapted to each context. Chapter 15 |
| **Heavyweight team**<br>Heavyweight team | A team with a leader who holds strong authority and dedicated members, suited to innovation that requires integration across units. Chapter 9 |

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| **Implementation climate**<br>Implementation climate | People's perception that using the innovation is expected, supported and rewarded. Chapter 15 |
| **Incremental innovation**<br>Incremental innovation | Innovation that builds on an existing offering or capability without changing its fundamental nature. Chapter 1 |
| **Influence without authority**<br>Influence without authority | A person's ability to move people who do not report to them, through reciprocity and partnership building. Chapter 6 |
| **Innovation**<br>Innovation | A new or significantly improved application that is adopted by users and creates value. Chapter 1 |
| **Innovation ambassador network**<br>Innovation ambassador network | People in different units who help colleagues articulate their ideas and connect the front line with the innovation system. Chapter 9 |
| **Innovation campaign**<br>Innovation campaign | A time-bound call for ideas on a specific challenge question, with a published mechanism for responding. Chapter 9 |
| **Innovation capability**<br>Innovation capability | An organization's ability to innovate repeatedly without depending on particular individuals. Chapter 6 |
| **Innovation implementation**<br>Innovation implementation | The consistent, actual use of an innovation by its intended users after the decision to adopt it. Chapter 15 |
| **Innovation inputs / outputs**<br>Innovation inputs / outputs | Inputs are the resources and conditions that enable innovation; outputs are its knowledge, technological and creative results. Chapter 2 |

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| **Innovation lab**<br>Innovation lab | A unit dedicated to experimentation and to tackling complex problems with new methods. Chapter 6 |
| **Innovation management**<br>Innovation management | The practices and capabilities that steer the flow of opportunities, ideas and initiatives, and manage experimentation, decisions and resources through to the realization of value. Chapter 1 |
| **Innovation management system**<br>Innovation management system | The formal framework through which an organization manages the flow of opportunities and initiatives through to value, encompassing strategy, governance, roles, resources and measurement. Chapter 5 |
| **Innovation process**<br>Innovation process | The successive stages an initiative passes through, from opportunity to implemented solution. Chapter 5 |
| **Innovation strategy**<br>Innovation strategy | A set of interlinked choices that define why an organization innovates, where, with what ambition and by what means. Chapter 4 |
| **Innovation theater**<br>Innovation theater | Activities that resemble innovation in appearance without bringing about change in products, services or ways of working. Chapter 1 |
| **Institutionalization**<br>Institutionalization | Embedding an initiative in the organization's routine systems so that it endures without relying on the enthusiasm of individuals. Chapter 20 |
| **Intelligent failure**<br>Intelligent failure | An unexpected result of a small, well-designed experiment in uncharted territory that yields new knowledge. Chapter 8 |

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| **Internal realities**<br>Internal realities | The organization's current state in terms of leadership, culture, structures, roles and skills. Chapter 2 |
| **Intrapreneurship**<br>Intrapreneurship | Entrepreneurial behavior by individuals within an existing organization to lead a new innovation. Chapter 1 |
| **Intrinsic task motivation**<br>Intrinsic task motivation | Being drawn to work by its interest and challenge rather than by an external reward. Chapter 7 |
| **Isomorphic mimicry**<br>Isomorphic mimicry | Adopting the forms of successful organizations without the capabilities that make them work. Chapter 20 |
| **Issue selling**<br>Issue selling | Employees' efforts to draw senior management's attention to issues they consider important and to shape how management understands them. Chapter 6 |
| **Job-relevant diversity**<br>Job-relevant diversity | Differences among team members in disciplines and expertise relevant to the problem. Chapter 9 |
| **Jobs to be done**<br>Jobs to be done | The progress a person seeks in a particular circumstance, enlisting a product or service to achieve it. Chapter 11 |
| **Lead users**<br>Lead users | People who face a need before others and expect large benefits from meeting it, and so innovate solutions for themselves. Chapter 11 |
| **Loose coupling**<br>Loose coupling | The condition of units that respond to one another while each retains its independence, so that what |

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| | happens in one carries over only weakly to the others. Chapter 20 |
| **Maturity model**<br>Maturity model | A description of successive levels of capability, used to assess the current position and identify the next step. Chapter 16 |
| **Micro-credential**<br>Micro-credential | Formal recognition of specific competencies acquired in a short learning program. Chapter 20 |
| **Mimetic isomorphism**<br>Mimetic isomorphism | The tendency of organizations, under ambiguity, to imitate organizations they see as successful or legitimate. Chapter 20 |
| **Minimum viable product**<br>Minimum viable product | The smallest version of a product that allows the greatest amount of learning with the least effort. Chapter 14 |
| **Mission-oriented innovation**<br>Mission-oriented innovation | Mobilizing efforts and resources toward a specific, ambitious outcome. Chapter 4 |
| **National innovation system**<br>National innovation system | The network of public and private institutions whose interactions produce, diffuse and use knowledge in a country. Chapter 18 |
| **Not-invented-here syndrome**<br>Not-invented-here syndrome | Resistance to ideas and knowledge that come from outside the team or the organization. Chapter 18 |
| **Open innovation**<br>Open innovation | A distributed innovation process based on knowledge flows purposively managed across organizational boundaries. Chapter 18 |

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| **Optimism bias**<br>Optimism bias | The tendency, when planning, to underestimate costs and risks and to overstate benefits. Chapter 17 |
| **Orchestrating unit**<br>Orchestrating unit | A unit linked to top leadership that connects the components of the existing system without owning their initiatives. Chapter 6 |
| **Organizational ambidexterity**<br>Organizational ambidexterity | An organization's ability to combine exploitation and exploration at the same time. Chapter 3 |
| **Organizational culture**<br>Organizational culture | The shared assumptions that a group has learned and passes on to new members as the correct way to think and act. Chapter 8 |
| **Perceived organizational support**<br>Perceived organizational support | An employee's belief that the organization values their contribution and cares about their well-being. Chapter 3 |
| **Pilot**<br>Pilot | A limited run of the solution in a near-real setting to test its value and adoption before scaling. Chapters 5 and 14 |
| **Problem finding**<br>Problem finding | The process by which a person identifies and formulates a problem before solving it; it is considered part of creativity. Chapter 11 |
| **Problem-driven iterative adaptation**<br>Problem-driven iterative adaptation | An approach that builds solutions locally, starting from problems, through experimentation, learning and continuous adaptation. Chapter 20 |
| **Procedural justice**<br>Procedural justice | The fairness of the way a decision is made, grounded in involvement, explanation and clear |

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| | expectations. Chapter 10 |
| **Production blocking**<br>Production blocking | The drop in idea generation in groups because members must wait their turn to speak. Chapter 12 |
| **Prototype**<br>Prototype | A representation of a solution built to answer a specific question about its role, its look and feel, or its implementation. Chapter 14 |
| **Proxy measure**<br>Proxy measure | A measure used in place of the intended construct because the latter is hard to measure directly; it may carry hidden biases. Chapter 19 |
| **Psychological safety**<br>Psychological safety | A shared belief within a team that asking questions, raising objections and admitting mistakes will not bring punishment or embarrassment. Chapter 8 |
| **Public value**<br>Public value | What society and its members value in the outcomes of public services and policies. Chapters 1 and 13 |
| **Randomized controlled trial**<br>Randomized controlled trial | An experiment in which participants are randomly assigned to the solution or an alternative, in order to isolate the solution's effect. Chapter 14 |
| **Real options**<br>Real options | A logic that treats a small early investment as the right to scale up later or to stop with a limited loss. Chapter 10 |
| **Reference class forecasting**<br>Reference class forecasting | Estimating cost and duration by looking at what actually happened in similar projects. Chapter 17 |

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| **Reframing**<br>Reframing | Formulating a problem from a different angle that reveals new solutions. Chapter 11 |
| **Relevance**<br>Relevance | The continuing connection between what an organization offers and the needs of its beneficiaries, now and in the future. Chapter 2 |
| **Responsible innovation**<br>Responsible innovation | Taking care of the future through collective stewardship of science and innovation in the present, grounded in anticipation, reflexivity, inclusion and responsiveness. Chapter 19 |
| **Results chain / logic model**<br>Results chain / logic model | A sequence linking inputs and activities to outputs, outcomes and impact. Chapter 16 |
| **Scenario planning**<br>Scenario planning | Sketching several plausible futures in order to test decisions against them. Chapter 4 |
| **Search field / Innovation arena**<br>Search field / Innovation arena | A priority area of opportunity that directs exploration efforts. Chapter 4 |
| **Servant leadership**<br>Servant leadership | A leadership style that makes serving followers and meeting their needs and growth an end in itself. Chapter 3 |
| **Solution requirements**<br>Solution requirements | What any solution must achieve to count as successful, regardless of its form. Chapter 12 |
| **Spending efficiency**<br>Spending efficiency | Achieving the greatest possible value with the available resources. Chapter 17 |

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| **Stage-Gate**<br>Stage-Gate | A system that divides an initiative's journey into stages separated by decision points with predefined criteria. Chapter 10 |
| **Staged funding**<br>Staged funding | Funding an initiative stage by stage, with each tranche tied to passing a gate. Chapter 10 |
| **Stages of concern**<br>Stages of concern | The progression of adopters' questions and concerns from the self, to the task, to impact. Chapter 15 |
| **Strategic triangle**<br>Strategic triangle | A framework that links the success of a public initiative to the alignment of public value, legitimacy and support, and operational capacity. Chapter 13 |
| **Student innovator journey**<br>Student innovator journey | An extended path that connects the university's units around what the student needs at each stage. Chapter 20 |
| **Sustainability-oriented innovation**<br>Sustainability-oriented innovation | Intentional changes to an organization's values, products and processes that create environmental and social value alongside economic value. Chapter 19 |
| **The innovator's path**<br>The innovator's path | A model that reads an idea's journey from its owner's perspective, from thought to practice, together with its outcomes and its surrounding environment. Chapter 7 |
| **Three horizons**<br>Three horizons | Dividing initiatives according to their distance from the current business and their degree of uncertainty. Chapter 4 |

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| **Total cost of ownership**<br>Total cost of ownership | The cost of a solution over its whole life, from purchase through operation to retirement. Chapter 17 |
| **Triple helix**<br>Triple helix | A model that explains innovation through the interaction of university, industry and government. Chapter 18 |
| **Unintended consequences**<br>Unintended consequences | Consequences of purposive action that the actor neither intended nor foresaw. Chapter 19 |
| **Value proposition**<br>Value proposition | The job a solution helps people get done, and why it is better than their alternatives. Chapter 13 |
| **Vanity metrics**<br>Vanity metrics | Numbers that look good in reports but do not indicate the value actually achieved. Chapter 16 |
| **Voltage effect**<br>Voltage effect | The decline in a program's impact as it scales from a pilot to wide implementation. Chapter 15 |
| **VUCA**<br>VUCA | A description of the environment along four dimensions: volatility, uncertainty, complexity and ambiguity. Chapter 2 |
| **Weak signals**<br>Weak signals | Early, ambiguous information about an approaching change. Chapter 4 |
| **Wicked problem**<br>Wicked problem | A problem that cannot be definitively formulated and that changes with every attempt to solve it. Chapter 11 |
| **Work environment for creativity** | The enablers and obstacles within an organization that affect employees' creativity, particularly their motivation. Chapter 7 |

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| Work environment for creativity | |
| **Workaround**<br>Workaround | A method people devise to cope with a problem that no one has solved for them. Chapter 11 |

## How does an organization move from scattered initiatives that celebrate innovation to a sustained capability to produce it?

This book traces the path of innovation from the first signal to a solution adopted in practice that creates value, and analyzes the organizational foundations that make this path a settled practice: strategy, systems, culture, governance, teams and measurement, with people at the center.

It rests on three pillars: a clear, high-priority problem; motivated people whose needs are met; and a system that connects them and turns effort into repeatable results. It combines international research evidence with the Saudi context, and pairs each chapter with a tool that can be used in a real organization.

Its twenty chapters are organized in five parts: foundations; institutional foundations and people; the initiative journey; value and the ecosystem; and an application to the university innovation ecosystem.